

Fraud Detection: Insights from Different Machine Learning Approaches

Data 698 Final Paper

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Abstract

Online payment fraud is a growing problem as more people rely on digital financial services. This study compares different machine learning models to evaluate how effectively they can detect fraud using a real-world dataset with over 6 million transactions. Because fraud cases make up less than 1% of the data, we used SMOTE to balance the training set and tested models on both the balanced and original imbalanced versions. The models included Logistic Regression, Naive Bayes, Decision Tree, Random Forest, SVM, and Neural Network. We focused on performance metrics like F1 Score and AUC, which give a clearer picture than accuracy in imbalanced settings. Overall, Random Forest and Neural Networks delivered the best results for detecting fraud, especially when balancing strong recall and precision.

Keyword

Fraud Detection, Machine Learning, SMOTE, Random Forest, Neural Network, Class Imbalance, AUC, F1 Score, Logistic Regression, Online Payment

Introduction

Online fraud is a growing problem, especially with the rise of digital payments and financial services. Every day, millions of transactions happen online, and while most are legitimate, some are fraudulent. Catching these fraudulent transactions quickly is important for protecting both consumers and businesses from financial loss.

Machine learning is a popular way to tackle this problem because it helps identify unusual patterns in large datasets—something that would be nearly impossible to do manually. Instead of using strict rules, machine learning models learn from past data to predict which transactions might be suspicious.

This study compares several machine learning models for detecting fraud in online payments. Instead of trying to find the single "best" model, the focus is on how each model performs in terms of accuracy, fraud detection, and how easy it might be to use in a real system. The analysis is based on a large real-world dataset with millions of transactions, providing a solid test for the models.

Literature Review

There are numerous studies that explore fraud detection using machine learning models. Many machine learning models, including Logistic Regression, Random Forest, and Neural Networks, have been evaluated and compared with each other to determine the most suitable model for fraud detection.

As per Almazroi & Ayub (2023), the study explores machine learning techniques applied to fraud detection and introduces the ResNeXt-embedded Gated Recurrent Unit (GRU) model (RXT). This model is designed to process real-time financial transaction data and is fine-tuned using the Jaya optimization algorithm. The study addresses key challenges in fraud detection, including data imbalance, feature extraction, and computational efficiency. The study incorporates techniques like SMOTE for balancing datasets, Principal Component Analysis (PCA) and Isolation Forest Model (IFM) for feature selection, and Autoencoder and ResNet-based feature extraction (EARN) for improved representation learning, which allow the model to capture both high-dimensional and low-dimensional transaction characteristics. The ResNeXt-GRU model combines ResNeXt for feature extraction and GRU for temporal pattern recognition, allowing the system to detect anomalies more effectively. This hybrid approach enables the model to recognize subtle patterns in fraudulent transactions, significantly improving fraud detection performance. Findings indicate that ResNeXt-GRU outperforms traditional ML models like Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM), achieving a 10–18% increase in classification accuracy while maintaining computational efficiency.

A study conducted by Pachhala et al. (2023) explores Random Forest, SVM, and Decision Trees for fraud detection, with Random Forest performing the best due to its ability to handle high-dimensional data and prevent overfitting. Their methodology included handling missing values, normalizing features, and encoding categorical variables, followed by feature selection techniques like correlation analysis and Recursive Feature Elimination (RFE). The study assessed the performance of the models using accuracy, precision, recall, and F1-score, Random Forest achieved an impressive 99.6% accuracy, surpassing other models. However, the study acknowledged several limitations for Random Forest, such as the lack of deep learning model comparison, no real-time fraud detection assessment, and limited handling of imbalanced data beyond standard preprocessing techniques.

Keerthi & Nalini (2024) explored hybrid models combining Random Forest, Logistic Regression, and Neural Networks for fraud detection. Unlike traditional models, their approach integrated real-time data streams and behavioral biometrics, improving fraud detection accuracy. Additionally, their study emphasized anomaly detection techniques, which were not explored in other research. The hybrid Random Forest and Neural Network combination performed best, achieving high accuracy with reduced false positives. Their study also incorporated 10-fold cross-validation to assess model performance, demonstrating that feature selection and real-time

anomaly detection significantly improved fraud detection accuracy. However, their work did not emphasize model interpretability, unlike other studies.

Overall, these studies highlight distinct ML approaches for fraud detection. Tree-based models like Random Forest balance accuracy and interpretability, while deep learning models such as RXT-GRU provide superior performance for complex fraud detection at the cost of computational efficiency. Hybrid models integrating multiple ML techniques, feature engineering, and real-time anomaly detection offer a promising solution for improving fraud detection systems.

Building on these insights, this study compares a wide range of traditional and advanced machine learning models, with special attention to class imbalance, interpretability, and evaluation using multiple performance metrics. Unlike some prior studies that focus narrowly on accuracy or omit deep learning benchmarks, this work evaluates both classical and neural models across SMOTE-balanced and imbalanced data to better reflect real-world conditions.

Research Question

How do different machine learning models—Logistic Regression, Random Forest, SVM, Decision Trees, Naive Bayes, and Neural Networks—compare in terms of performance for detecting fraud in online payment transactions?

Data Description and Acquisition

The dataset used in this study was sourced from Kaggle and contains detailed information on 6,362,620 online payment transactions. It includes 11 variables that describe the context of each transaction, such as the transaction type, amount, balances before and after the transaction, and whether it was marked as fraudulent (isFraud). This dataset offers a realistic and large-scale view of online financial behavior, making it ideal for testing fraud detection models.

The response variable is isFraud, a binary indicator where 1 represents a fraudulent transaction and 0 represents a legitimate one. Predictor variables include transaction metadata such as amount, account balances, and transaction types. Additional features were engineered to better capture transaction behavior, including the difference between sender and receiver balances and robustly scaled versions of continuous variables.

Several preprocessing steps were performed to ensure statistical validity:

- Variable types were verified, and isFraud and type were converted to factor.
- Transactions with zero amount were removed, as they have no meaning in fraud detection.
- Numeric and categorical variables were visualized to explore distribution and skewness.
- Log transformation was applied to highly skewed features (e.g., balances).
- Outlier handling used robust scaling based on median and IQR.
- One-hot encoding was applied to convert the type variable into multiple binary columns.
- New features were created to reflect balance differences.

- Final feature selection was performed using correlation analysis and Variance Inflation Factor (VIF).

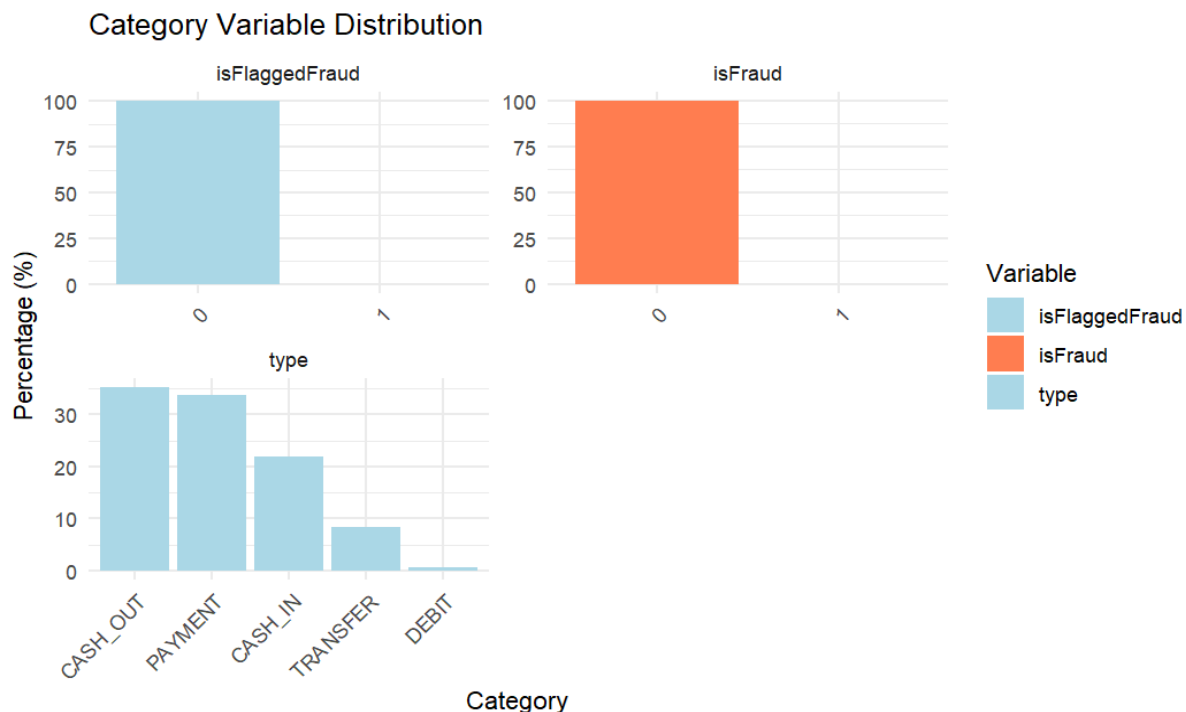
Due to a significant class imbalance (fraudulent transactions represent less than 1% of the dataset), the SMOTE (Synthetic Minority Oversampling Technique) algorithm was applied to the training data. This ensures that models are not biased toward predicting the majority (non-fraudulent) class and allows for more balanced evaluation of performance.

Exploratory Data Analysis

Category Variables Distribution

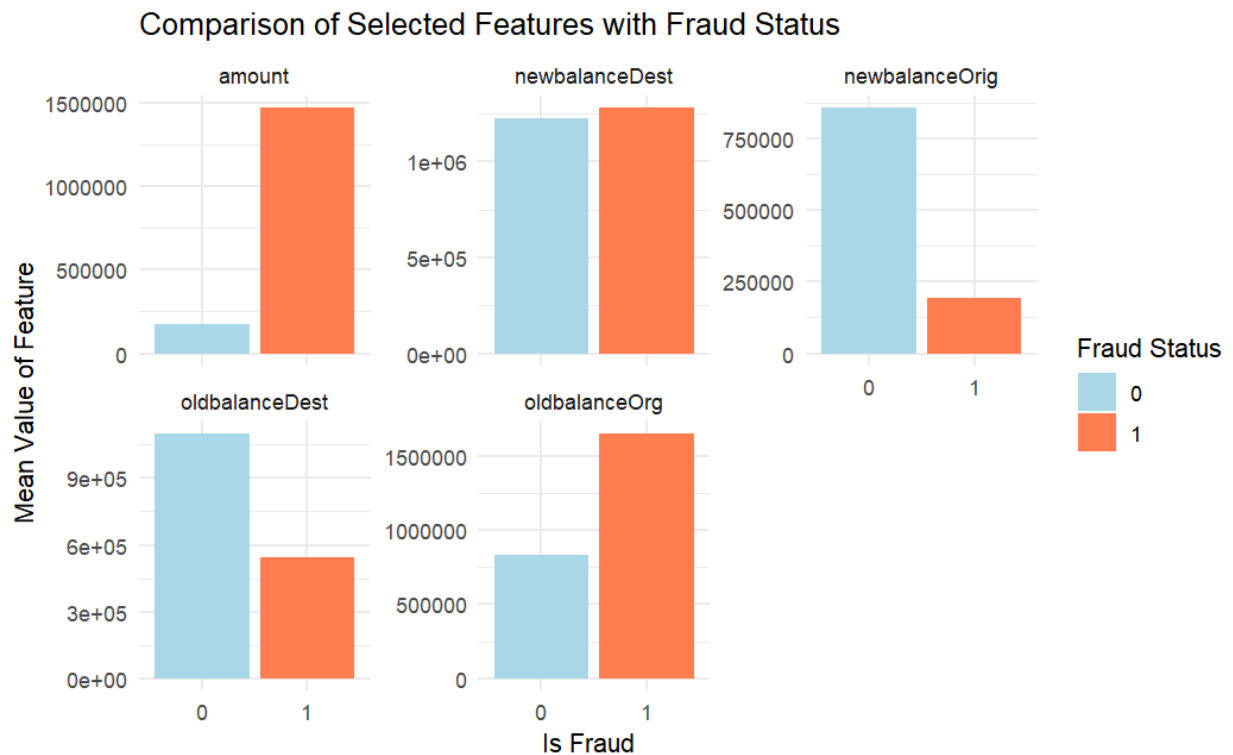
The isFraud column is highly imbalanced, with around 99% of transactions being legitimate and fewer than 1% classified as fraudulent. This reflects typical real-world fraud detection challenges. To preserve natural patterns, the imbalance was not corrected during EDA; it was addressed later during modeling using techniques such as SMOTE and downsampling.

The type variable is also skewed, with most transactions falling into the CASH_OUT and PAYMENT categories. Although other types like DEBIT appear less frequently, no changes were made during EDA. Notably, fraud is most commonly seen in the CASH_OUT and TRANSFER transaction types.

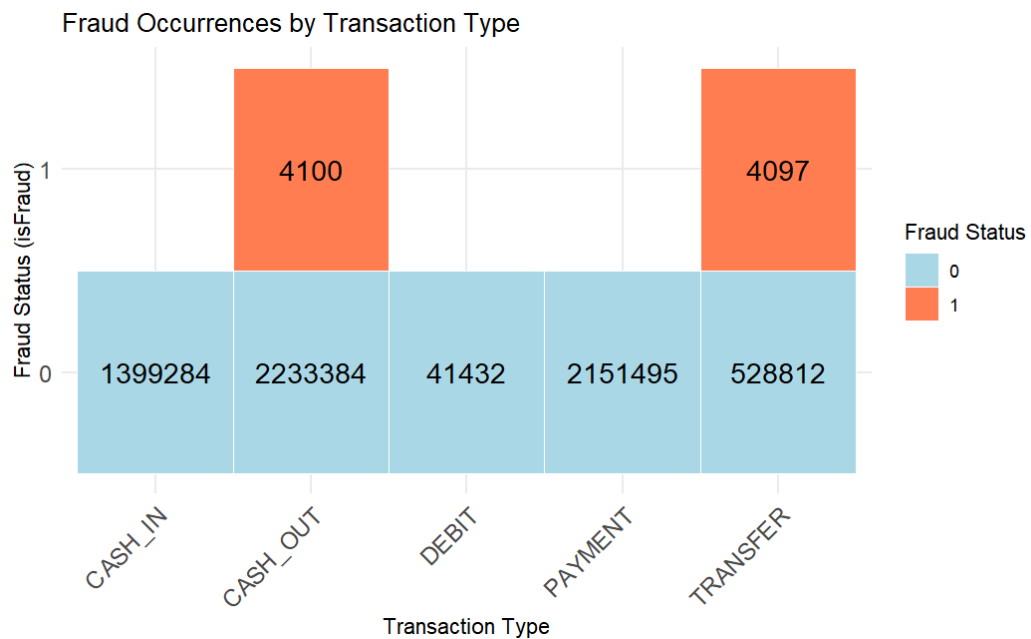


Features Observations

Fraudulent transactions tend to involve significantly larger amounts compared to normal ones (amount), suggesting that fraudsters often target high-value transfers. Accounts used to initiate fraud usually have higher original balances (oldbalanceOrg), making it easier to move large sums. After a fraudulent transaction, these origin accounts (newbalanceOrig) are often left with little or no balance—indicating that fraudsters are draining them completely. On the receiving side, destination accounts (oldbalanceDest) involved in fraud typically start with low balances, possibly to avoid detection or because they're newly created. Interestingly, the average ending balance in destination accounts (newbalanceDest) is similar for both fraudulent and non-fraudulent transactions, which may suggest that balance restoration isn't a strong indicator of fraud by itself.

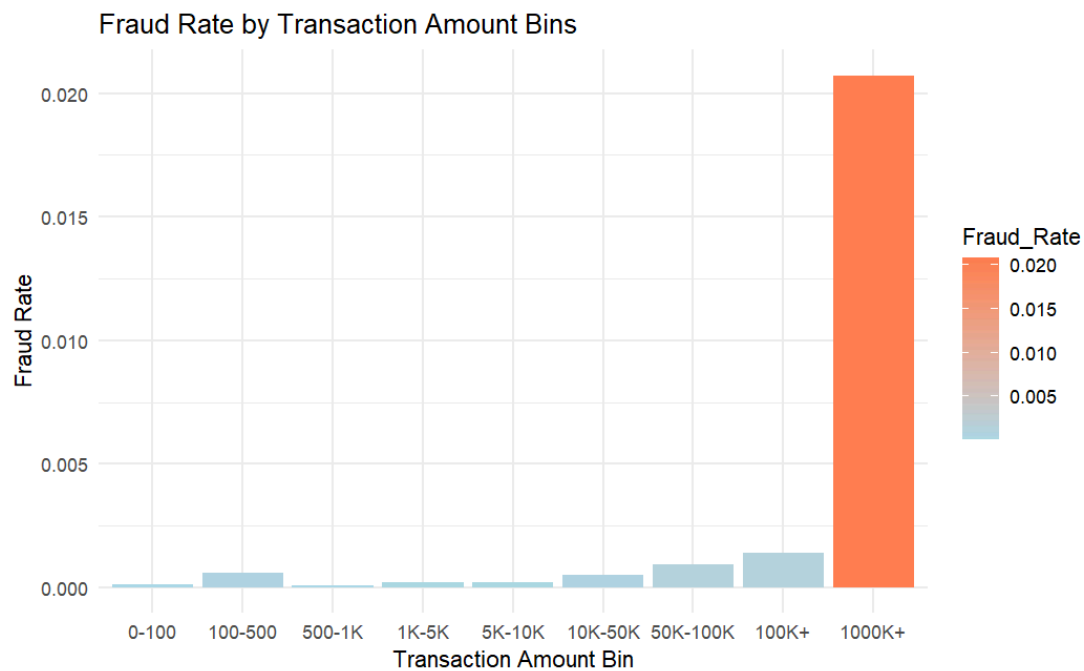


Heatmap analysis showed that fraudulent activity is concentrated in CASH_OUT and TRANSFER transaction types.



Fraud by Amount Range

Fraud was more frequent in transactions exceeding \$1,000,000. This suggests that higher-value transactions are more vulnerable and attractive targets for fraud.



Statistical Methods

After preprocessing and cleaning the data, the dataset was split into training and testing sets. To address the severe class imbalance (less than 1% of transactions are fraud), the SMOTE algorithm was used to oversample the minority class. Then, to prevent overfitting from synthetic examples and better simulate real-world proportions, the majority class was downsampled to create a final training set where non-fraudulent cases were approximately twice as many as fraudulent cases.

Six machine learning models were trained and evaluated:

- **Logistic Regression:** A linear model used for binary classification, known for its simplicity and interpretability.
- **Naïve Bayes:** A probabilistic model based on Bayes' Theorem, suitable for high-dimensional data.
- **Decision Tree:** A flowchart-like model that splits data based on feature values; useful for understanding decision rules.
- **Random Forest:** An ensemble of decision trees that improves accuracy and reduces overfitting.
- **Support Vector Machine (SVM):** A model that finds an optimal hyperplane to separate classes; effective for complex boundaries.
- **Neural Network (using Pytorch):** A deep learning model that captures complex, non-linear patterns; powerful but less interpretable.

In fraud detection tasks with imbalanced datasets, accuracy alone can be misleading. A model could achieve over 99% accuracy just by predicting every transaction as non-fraud, but it wouldn't actually catch any fraud cases. That's why more focus was placed on metrics like precision, recall, F1 Score, and AUC. Recall shows how many fraud cases the model catches, while precision tells how many of the flagged transactions are actually fraudulent. The F1 Score balances these two, offering a better sense of overall model performance. AUC gives an overall measure of how well the model separates fraud from non-fraud. These metrics give a much clearer picture of how effective each model is at detecting fraud. In general, higher values for precision, recall, F1 Score, and AUC indicate better performance. A strong fraud detection model will ideally have both high recall (to catch as many fraud cases as possible) and high precision (to reduce false alarms), which results in a higher F1 Score and a more trustworthy model overall.

Model selection was guided by several factors:

- Logistic Regression and Naive Bayes offer baseline interpretability and simplicity.
- Tree-based models (Decision Tree, Random Forest) handle nonlinear interactions and variable importance well.
- SVM is effective for classification in high-dimensional settings.

- Neural Networks were included to test performance on nonlinear patterns in large datasets.

To provide a well-rounded comparison, Decision Tree, Random Forest, and Neural Network models were also trained and evaluated using the original imbalanced dataset in addition to the SMOTE-balanced version. This helped assess how each model handled real-world class distributions and whether performance gains were dependent on synthetic oversampling.

These models provide a range of interpretability, scalability, and complexity, allowing for a well-rounded comparison across different fraud detection criteria.

Experimentation and Results

Model Performance Summary (Balanced + Imbalanced):					
Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression (SMOTE)	0.9679	0.0379	0.9799	0.0729	0.9947
Naive Bayes (SMOTE)	0.9601	0.0262	0.8279	0.0507	0.9768
Decision Tree (SMOTE)	0.9995	0.7132	0.9725	0.8229	0.986
Random Forest (SMOTE)	0.9997	0.9609	0.809	0.8784	0.9865
SVM (SMOTE, SGDClassifier)	0.9312	0.018	0.9762	0.0353	0.9917
Neural Network (SMOTE)	0.9973	0.3193	0.9957	0.4836	0.9995
Decision Tree (Imbalanced)	0.9997	0.9057	0.8615	0.8831	0.9307
Random Forest (Imbalanced)	0.9997	0.9579	0.8048	0.8747	0.9883
Neural Network (Imbalanced)	0.9996	0.9505	0.7383	0.831	0.9955

Models trained on the SMOTE-balanced dataset showed the following results:

- **Logistic Regression (SMOTE):** Caught almost all the fraud (recall = 0.9799), but had very low precision (0.0379), meaning it flagged a lot of legit transactions too. The F1 Score (0.0729) was low, even though AUC was strong at 0.9947.
- **Naive Bayes (SMOTE):** Similar pattern to Logistic Regression. High recall (0.8279), low precision (0.0262), and F1 Score (0.0507). AUC came in at 0.9768.
- **Decision Tree (SMOTE):** Showed better balance. Precision (0.7132) and recall (0.9725) led to a strong F1 Score (0.8229). A solid performer that caught fraud while minimizing false alarms.
- **Random Forest (SMOTE):** It consistently caught fraud while keeping false alarms low. High precision (0.9609), good recall (0.8090), and the best F1 Score (0.8784). AUC was also strong at 0.9865.
- **SVM (SMOTE):** Very high recall (0.9762), so it found nearly all fraud cases. But

precision (0.0180) and F1 Score (0.0353) were very low. The AUC, based on the decision function, was solid at 0.9917.

- **Neural Network (SMOTE):** Outstanding recall (0.9957) and better precision (0.3193) than Logistic or Naive Bayes. F1 Score (0.4836) and the best AUC (0.9995) among all models.

Models trained on the imbalanced data performed surprisingly well:

- **Decision Tree (Imbalanced):** Did well even without data balancing. High F1 Score (0.8831), precision (0.9057), and recall (0.8615).
- **Random Forest (Imbalanced):** Again, solid performance with F1 Score (0.8747), precision (0.9579), recall (0.8048), and AUC (0.9883).
- **Neural Network (Imbalanced):** Slightly lower recall (0.7383) than the SMOTE version, but higher precision (0.9505) and still strong F1 Score (0.8310). AUC was 0.9955.

Overall, Random Forest stood out as the most reliable model across both balanced and imbalanced datasets. Neural Networks were also strong performers, especially in terms of recall and AUC. Logistic Regression and Naive Bayes were better for understanding the data, but didn't do as well at accurately detecting fraud.

Discussion and Conclusions

The results show that each machine learning model has its own strengths, and the best one depends on what the goal is—whether it's maximizing fraud detection, reducing false positives, or understanding how different features impact fraud.

- **Logistic Regression and Naive Bayes** are simple and easy to explain. Logistic Regression especially is helpful when the goal is to understand which factors—like transaction amount or account balance—are linked to fraud. But while these models had great recall (they found most of the fraud cases), their precision was extremely low, meaning they also flagged a lot of legitimate transactions. This led to low F1 scores, which makes them less ideal when false alarms are a concern.
- **Decision Tree and Random Forest** models performed much better. Random Forest stood out as one of the top performers with strong results on both SMOTE-balanced and imbalanced datasets. It combined high recall and precision, which means it was accurate and consistent. Decision Trees also did well, especially on the imbalanced data, where they had the highest F1 score. On top of that, these models are easier to interpret than deep learning approaches, which can be useful when explaining results to business users or stakeholders.
- **Neural Networks** had the highest recall overall, especially when trained on SMOTE-balanced data. This means they were great at catching fraud. However, their

precision was lower in that setting, so they had more false positives. Interestingly, when trained on the original imbalanced dataset, the F1 score improved—suggesting that sometimes Neural Networks perform better when trained on real, unbalanced data instead of synthetic samples. They take longer to train and are harder to interpret, but they're good at learning complex patterns.

- **SVM (using SGDClassifier)** also had excellent recall but very low precision. That means it caught almost every fraud case but also misclassified a lot of legit ones. Still, when using the decision function for evaluation, the AUC came out high (0.9917), showing the model was still making useful distinctions between fraud and non-fraud.

If the goal is to catch fraud effectively, Random Forest and Neural Networks are the best picks based on both F1 Score and AUC. They handled the imbalanced data well and picked up on complex fraud patterns. On the other hand, if the priority is to understand what's driving fraud, then Logistic Regression is still valuable thanks to its transparency and interpretability.

Ethical Considerations

All data used were anonymized and publicly available, avoiding any personally identifiable information. To ensure fairness and reduce bias, SMOTE and downsampling were applied to mitigate class imbalance. Sensitive or identity-based variables were excluded from modeling. Feature engineering focused on transaction behavior rather than personal attributes. Models were evaluated using metrics that reflect both false positives and false negatives. This approach aligns with ethical practices in machine learning by prioritizing data privacy, minimizing bias, and promoting transparent, responsible model evaluation.

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